

Inductor-free LoRa Backscatter

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Abstract—LoRa backscatter achieves long-range communication at the cost of only tens of micro-watts of power when implemented in integrated circuits (ICs), which makes it a potential enabler for massive IoT. However, despite the above advantages, we find that the current tag design wastes approximately 72% of the chip functional area due to the use of large-size inductors in the impedance loads array. This inefficiency significantly increases the cost per chip during mass production.

To address this die area issue, we propose OsTAG, a novel LoRa backscatter design that eliminates all inductors in the impedance loads array while maintaining the same quantization resolution. The key innovation lies in the creation of virtual impedance using oversampling. Realizing such design requires overcoming the technical challenges of mitigating approximation error and managing oversampling clock imperfections. To validate our design, we develop prototype and an IC, demonstrating promising results in terms of both performance and efficiency. OsTAG is expected to cost only 28% of the existing chip area while achieving comparable hundred-meter level communication range.

Index Terms—Backscatter, LoRa.

I. INTRODUCTION

Backscatter communication [2], [6], [9], [11], [15], [19], [20], [27], [29], [31], [34], [38]–[42], which provides dual-benefits of energy harvesting and μW -level power consumption, has emerged as an alternative to active radios in the area of Internet of Things (IoT). Among recent advancements, LoRa backscatter [31] has made significant progress in overcoming the traditional 1 m to 10 m communication range limitation. It extends this range to 100 m to 1000 m, which is crucial for applications such as industrial sensor networks, forest monitoring, and building health monitoring—key use cases highlighted in the 3GPP Ambient IoT technical report [1]. By leveraging Chirp Spread Spectrum (CSS) modulation, LoRa backscatter achieves this extended range, providing a robust solution for large-scale IoT deployments. In contrast, Wi-Fi backscatter systems can only achieve a limited range within 10 m. This makes it a potential technique enabler for massive IoT, supporting applications such as smart logistics, smart agriculture, and industry sensor network.

However, once the network reaches an unprecedented scale, the cost of each tag will become a decisive factor for the real-world deployment of LoRa backscatter. Taking logistic

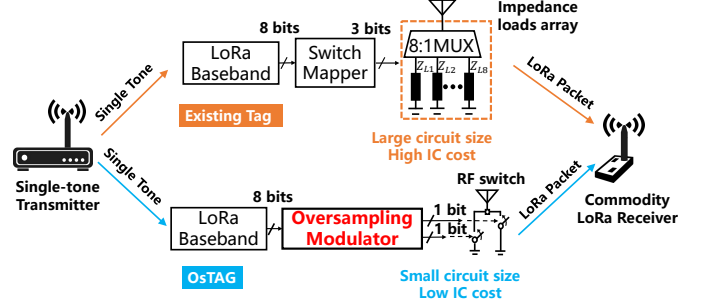


Fig. 1: The problem of traditional LoRa backscatter tag: it requires large-size inductors as impedance loads that increase IC cost. In contrast, the OsTAG design does not use inductors and has a small size.

network (the main battleground of the backscatter technique) as an example: companies like Alibaba, Amazon or Walmart need to handle a package volume that is over 87 billion per year. That is to say, even a 1 cent increase in the cost of each tag can bring over 870 million additional cost to the company. However, it is difficult to shrink down the cost of a LoRa backscatter, because the current design of LoRa backscatter tag cannot be easily integrated into a small-form chip. Since the size of each chip directly determines the number of chips obtained on a wafer, increasing the chip size means multiplying the cost of each chip.

So, why the existing LoRa backscatter cannot be integrated into a small-form chip? The answer is illustrated in Fig. 1. Specifically, to synthesize the complex CSS signals, the tag has to manipulate the amplitude and phase of backscatter signals in fine granularity. Since both these two parameters are determined by the tag's load impedance, we need to use an impedance array to achieve fine-grained load quantization. Existing LoRa backscatter tag typically uses a 8-element impedance load array to achieve 3-bit load quantization. However, one problem in this design is that each impedance requires an large-size inductor as the impedance load. Since the size of an inductor directly determines its inductance value, we cannot reduce its size even using finer IC process. For example, inductors operating on the 900MHz band should be at least 21 nH, which will incur an unavoidable 0.38 mm^2 area occupation. Considering that the baseband logic occupies only $100 \mu\text{m}^2$ under a 40nm IC process, the use of inductor brings an additional 72% of area occupation.

To overcome the above challenge, we present the design and implementation of OsTAG, an inductor-free LoRa backscatter design. The unique ability of OsTAG is to make a fine-grained control of the signal's amplitude and phase using only a 1-bit RF switch which flips simply between OFF and ON stages. It

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achieves this leveraging the fact that the LoRa receiver samples the signal by averaging the instantaneous power of the signal with a bandpass filter. So, if a tag can flip its stage with a much higher frequency than the receiver's ADC sampling rate, it can generate signal with different amplitudes by adjusting the duty-cycle of the generated pulse wave. Moreover, given that the phase of a signal can be manipulated by simultaneously changing its in-phase and quadrature amplitude, it is also feasible to manipulate complex signal with 1-bit RF switches.

Note that the essence of OsTAG is to transform the tag's ability in performing fine-grained time domain manipulation to the ability in performing fine-grained amplitude domain manipulation. This transformation yields the following two significant advantages:

- **Reduction in chip area and cost:** Fine-grained time domain manipulation does not need inductors and capacitors in the tag IC, which reduces the chip area by 72% compared to existing designs. This reduction in chip area can be roughly translated to cost reduction of 72% per chip. Although increasing tag's flipping frequency may incur additional energy consumption, we can mitigate it by using extremely low-power and area-efficient clock circuits such as free-running ring oscillators.
- **Scalable resolution support:** Our design allows for convenient scaling of quantization resolution up to 4 bits or higher without increasing circuit area. In contrast, existing methods require 16 physical LC components for a 4-bit impedance loads array and 32 for a 5-bit impedance loads array. Our design only requires 64 and 256 times the sampling clock without changes in the RF front end.

(a) Reducing approximation error in switching process:

To minimize the approximation error in the switching process, it is imperative for the modulator to meticulously choose the appropriate impedance state that would result in the desired virtual impedance. Additionally, the modulator must ensure that the harmonic compression effects remain consistent with the method that uses real impedance.

To solve this problem, we propose a purely digital modulator that uses a feedback structure to synthesize virtual impedance via switching. This process moves almost all approximation error to unwanted higher frequencies in real time. This method, which is called delta-sigma modulation, has the potential of realizing arbitrarily low approximation error under high oversampling clock rates.

(b) The imperfections of low-power oscillators: Low-power oscillators such as ring oscillators, operating at the μW -level and MHz frequency range, are known to be inaccurate in clock frequency [42] and unstable in edge-to-edge period (jitter). This can result in significant carrier frequency offset (CFO) and sampling frequency offset (SFO), which can prevent successful LoRa packet reception. To address this problem, we propose an asynchronous hybrid-clock framework design, where a low-frequency accurate clock and a high-frequency inaccurate clock to drive the baseband logic and the oversampling logic, respectively. This hybrid clock framework successfully avoids CFO and SFO caused by the unstable low-power clock while still benefiting from the oversampling design.

In addressing the challenges above, we make the following contributions:

- We for the first time propose an oversampling technique to synthesize a *virtual impedance*. The virtual impedance can function like true impedance while needing no physical inductors or capacitors in the circuit. This technique enables ultra-low cost and small form factor LoRa backscatter tag, pushing the real-world deployment of LoRa backscatter in massive IoT.
- We embed the above technique in a full system that overcomes additional practical challenges, including: i) timing misalignment between the tag and the receiver; ii) serious CFO and SFO brought by the low-power clock – all of which lead to an imperfect synthesis of the CSS signal.
- Thirdly, we design and implement OsTAG, an inductor-free LoRa backscatter system capable of achieving a commodity-compatible LoRa connection with a throughput of 0.3-5 kbps and a bi-directional communication range of 100-200 meters. When the tag is made into IC, it is estimated to save 72% chip area compared to existing designs. We prototype the whole circuits in a printed circuit board (PCB) measuring $4\text{ cm} \times 4.3\text{ cm}$ and conduct performance evaluations.

II. EXISTING LORA BACKSCATTER

We in this section introduce why the existing LoRa backscatter cannot be integrated on a small-sized chip. Specifically, we first introduce how existing LoRa backscatter synthesizes LoRa baseband signal by switching among local impedance and why the impedance loads array is a necessity for signal synthesis. Then we introduce the why such a design can incur the die area issue in IC process.

A. Introduction to LoRa Backscatter

LoRa Modulation Technique: LoRa employs the CSS modulation technique, which utilizes chirp signals as symbols. Each symbol can choose one out of 2^{SF} distinct chirps, thereby carrying SF bits. In this context, SF refers to the spread factor of CSS modulation, which partially determines the chirp slope. Different chirps can be generated by altering the starting frequency of a standard upchirp. The frequency of this standard upchirp linearly increases with time from $-BW/2$ to $BW/2$, where BW denotes the bandwidth. A discrete-time mathematical expression for a LoRa symbol can be represented as:

$$c[n] = e^{j2\pi(0.5kn^2 + \omega_0 n)}, \quad (1)$$

where n is the n^{th} sampling point, k denotes the chirp slope, and ω_0 represents the normalized starting frequency of the chirp. This digital baseband waveform is then converted to analog baseband waveform by a Digital-to-Analog Converter (DAC) and subsequently the radio front-end up-converts the baseband waveform to the carrier frequency and sends it out. The final RF signal is illustrated below:

$$y(t) = A \cdot \text{Re}[c(t) \cdot \cos(2\pi f_c t)], \quad (2)$$

where A is the amplitude after the Power Amplifier (PA), f_c is the carrier frequency, and $c(t) = e^{j2\pi(0.5kt^2 + \Omega_0 t)}$

is the continuous-time signal derived from $c[n]$ after the DAC's conversion, and $\Omega_0 = \omega_0/BW$ is the physical starting frequency of each chirp. By selecting different Ω_0 , the signal $y(t)$ can carry different data bits for the receiver.

LoRa Backscatter: Different from active LoRa transmitters, LoRa backscatter synthesizes a CSS signal without using a DAC, upconverter, or Power Amplifier (PA) in the RF frontend. Instead, it synthesizes CSS signals by changing the impedance load of the tag's antenna. Specifically, each sample on the CSS signal $c[n]$ can be mapped to a point on the I-Q plane, which can be generated by adjusting the phase and amplitude of the carrier signal, as shown in Fig. 2. Since the above two parameters can be further altered by changing the reflection coefficient of the antenna, then to generate the CSS signal, we can let the transmitter send a single tone to the LoRa backscatter tag, then the tag synthesizes the CSS signal $c[n]$ by switching to the impedance load with a reflection coefficient S_{11} that best approximates the current complex value of $c[n]$:

$$\arg \min_i D[n] = \|S_{11}(i) - c[n]\|. \quad (3)$$

However, it is inefficient to generate each sample with a specific impedance load, which means to use a large impedance loads array to generate all the CSS signals. To solve this problem, LoRa backscatter approximates the CSS signal by selecting 8 representative samples and map them to the I-Q plane. Then the signal can be generated by using a 8-element impedance loads array. Fig. 2 illustrates this approximation. The left graph displays the real part of a chirp signal. Each sample is approximated by the eight points on the right graph, which represent all possible S_{11} values provided by the LoRa backscatter impedance loads. Note that for illustration purposes, all the S_{11} values in Fig. 2 are located in the ideal positions ($e^{j*(0,0.125,\dots,0.875)*2\pi}$). In reality, however, the eight S_{11} values generated by the physical impedance loads can deviate slightly from the ideal positions due to non-ideal resistors, capacitors, and inductors.

Now, suppose the incident RF signal is $S_{in}(t) = A \cdot \cos(2\pi f_c t)$. Then, the tag generates the backscattered $S_{bs}(t)$ by changing its reflection coefficient S_{11} as $S_{bs}(t)$ satisfies $S_{bs}(t) = \text{Re}[S_{in}(t) \cdot S_{11}(t)]$. Since $S_{11}(t)$ is an approximated version of the LoRa baseband signal $c(t)$, we denote it as $c_a(t)$. We can then obtain:

$$S_{bs}(t) = A \cdot \text{Re}[c_a(t) \cdot \cos(2\pi f_c t)]. \quad (4)$$

Clearly, to make the synthesized signal c_a closer to the CSS signal c , we need to manipulate the signal c_a with a more fine-grained load quantization. Existing LoRa backscatter systems that adopt 3-bit quantization can already achieve satisfactory performance. Higher quantization resolution can make $S_{bs}(t)$ smoother and closer to the CSS signal, which further facilitates higher packet reception rate (PRR) and lower bit error rate (BER).

B. Inductor Challenge of LoRa Backscatter

Although increasing the number of impedance loads can offer more S_{11} stages and produces a synthesized signal

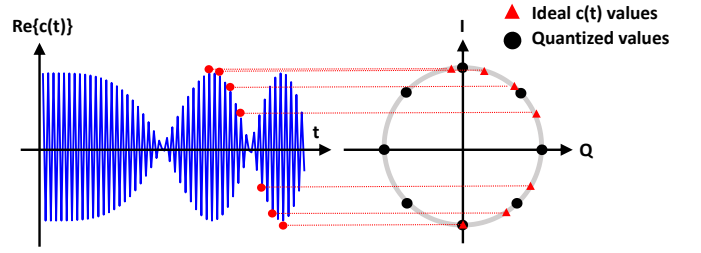


Fig. 2: Approximation of CSS Signal to nearest S_{11} Values. The real (red) and imaginary (blue) components of a chirp signal are depicted on the left, while their quantized counterparts are illustrated on the right.

$c_a(t)$ that is closer to $c(t)$, this also presents practical implementation challenges. Specifically, implementing an array of impedance loads incurs high costs, whether on a printed circuit board (PCB) or on an integrated circuit (IC). To fully understand this issue, we first examine the deployment of all impedance loads on a PCB, which is the most common method for circuit implementation.

Cost of impedance loads array on PCB: Placing impedance loads on a PCB requires a large dedicated area and poses the challenge of achieving accurate impedance values. The impedance loads used in existing backscatter designs are all discrete components with physical properties that determine their values. Suppose we want to place a 3-bit load array, we need to first consider what S_{11} values should be provided for a good quantization result. For the ideal case, we let these values be $e^{j*(0,0.125,\dots,0.875)*2\pi}$ as shown in Fig. 2. Next, we calculate each impedance $Z_L(i)$ for the load according to its S_{11} value $S_{11}(i)$, using the equation:

$$Z_L(i) = \frac{1 + S_{11}(i)}{1 - S_{11}(i)} \cdot Z_0, \quad (5)$$

where $Z_0 = 50\Omega$. By substituting S_{11} with the aforementioned ideal values, we can obtain all 8 values of Z_L as follows: $\infty, 120.7j, 50j, 20.7j, 0, -20.7j, -50j, -120.7j$. Note that the Z_L values with a positive imaginary part is realized with inductors ($Z_L = j2\pi fL$), whereas those with a negative imaginary part is implemented with capacitors ($Z_L = \frac{-j}{2\pi fC}$). Based on the relationship between L/C and their impedance, we can derive the inductance and capacitance values, respectively.

At every moment, the antenna is connected to only one load, and a 3-bit control word controls an 8:1 RF multiplexer to select among these impedance loads. This circuit-level switching is commonly achieved by cascading several levels of 4:1 RF multiplexers (also known as SP4T).

The above framework introduces two sources of costs:

(a) *Size cost.* Due to the size constraints of the RF multiplexers, passive impedance loads, and the RF transmission line (with length control) interconnecting these components, the circuit size is significantly larger compared to tags that only have two impedance loads [14], [29], [42].

(b) *Precision loss.* The discrete L and C components exhibit large deviations (typically 2%-20% [24], [25]); moreover, their values are not continuously available (one cannot purchase L and C components with special values). To address these issues

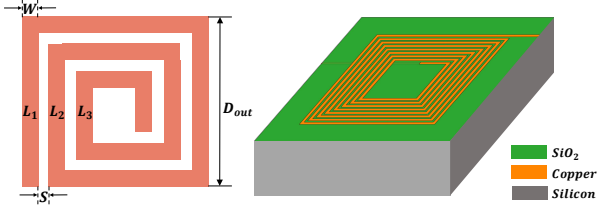


Fig. 3: The 2D model and the 3D structure of a spiral inductor in IC. The inductor, primarily composed of copper traces on a semiconductor substrate, derives its inductance from its distinctive geometric configuration.

and achieve a smaller tag size and more accurate impedance, it is essential to integrate the entire impedance loads into the backscatter IC.

Cost of an impedance loads array in IC (inductor challenge). Integrating the entire impedance loads array into the IC appears to be a promising solution, as it overcomes the limitations associated with PCBs: (a) The RF multiplexer is extremely small, as it consists of only a few transistors and occupies negligible area compared to the PCB design scheme. This is due to the fact that the chip is a lumped-parameter circuit with a scale ($\approx$ mm) much smaller than the sub-GHz wavelength λ (>30 cm). (b) The impedance loads can be designed with high precision. For instance, spiral inductors are widely used as on-chip inductors, and their basic structure is described in [28]. All impedance loads can be finely tuned by controlling the inductor length, the spacing between lines, the linewidth, and the total rounds. The detailed geometric shape is illustrated in the left of Fig. 3. The relation between the geometric parameters with the final inductance L is approximately given by:

$$L \approx \frac{1.3 \times 10^{-7} \cdot l_{\text{tot}}^{5/3}}{\left[\frac{l_{\text{tot}}}{4N} + W + (N-1)(W+S)\right]^{1/3} W^{0.083} (W+S)^{0.25}}, \quad (6)$$

Here, W represents the width of the spiral inductor, S denotes the spacing between adjacent spiral turns, and N indicates the total number of turns in the spiral inductor. The total length of the inductor line is approximated as $l_{\text{tot}} \approx 4N[D_{\text{out}} - W - (N-1)(W+S)]$. These parameters can be adjusted to design an inductor with high accuracy, typically through iterative tuning and RF simulation in HFSS.

However, we demonstrate that the IC cost of an impedance loads array remains high, primarily due to the large size of inductors. To demonstrate this issue, we utilize a straightforward model from [23] to estimate their size, as shown in Fig. 3's right side. We initially examine a single-layer on-chip inductor model. Recall that existing LoRa backscatter requires inductance to generate $Z_L = 120.7j, 50j, 20.7j$, which correspond to inductance of 21.3 nH, 8.8 nH, and 3.7 nH when the RF carrier is set to 900 MHz. We then design the corresponding inductors (assuming they can be implemented). The result shows that its size reaches approximately $400 \mu\text{m}$ with $N = 8$. When we use three inductors, they occupy a total area of $3.8 \times 10^5 \mu\text{m}^2$. This area is roughly 16.5 times the size of the LoRa digital baseband after converting RTL to layout (approximately $2.3 \times 10^4 \mu\text{m}^2$). In other words, producing one

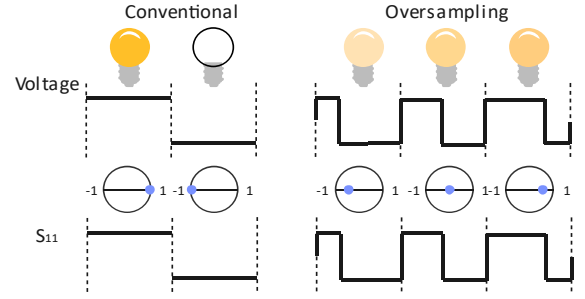


Fig. 4: Naive oversampling: analogous to employing PWM for regulating LED brightness, the tag control logic alternates between two impedance states to generate a desired virtual impedance for arbitrary S_{11} values.

LoRa backscatter chip could cost the same as producing 16.5 inductor-free chips.¹ The advantages of removing inductors and the impedance loads array are substantial.

III. NAIVE OVERSAMPLING

In this section, we first introduce a basic version of our oversampling method which tries to synthesize a virtual impedance to replace physical impedance loads for size reduction. Then, we analyze and discuss the technical challenges associated with this naive oversampling method.

A. Analysis of Inductor Removal

In our design, we emulate a virtual impedance using only a 1-bit RF switch which flips between two distinct impedance values: $Z_L = \infty$ for $S_{11} = e^{j0}$ and $Z_L = 0$ for $S_{11} = e^{j\pi}$. Note that both these two impedance values can be produced without capacitors or inductors, as both their S_{11} and Z_L imaginary components are zero. This makes it feasible to synthesize arbitrary impedance with an inductor-free design.

How to synthesize arbitrary Z_L with ∞ and 0 ? Our inspiration comes from the digital design of LED brightness control. Specifically, the brightness of an LED depends on the current through it. So, a straightforward solution is to generate a continuous control signal using an analog device to adjust the supply voltage for the LED. This method is however cost-prohibitive since it necessitate a DAC and a mapping logic for the voltage-brightness relation, which adds complexity and expense to the system. A more elegant solution is to use a digital signal to control brightness. Digital control circuits, like MCUs, can switch between ON and OFF states at high frequencies using pulse-width modulation (PWM). Then leveraging the visual residual impact on the human eye, we can adjust the brightness of LED by adjusting the percentage the LED is ON versus OFF. For example, by maintaining an ON state for 23% of a period and an OFF state for the remaining 77%, it will appear to the eye that the LED attain a 23% brightness.

¹Consider backscatter circuit area only. A more accurate calculation that also considers the area of other auxiliary analog circuits is given in Sec. VI-F.

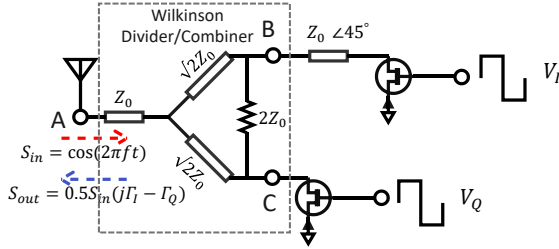


Fig. 5: In-phase and Quadrature modulator circuit that realizes complex arbitrary S_{11} using oversampling.

B. Real-valued S_{11} Synthesis

In the context of virtual impedance emulation, we can employ the PWM technique to rapidly alternate the value of Z_L between ∞ and 0, then a virtual impedance of any desired value can be emulated by adjusting the duty cycle of the generated PWM signal. This approach can be divided into two phases: (a) generating ∞ and 0 impedances, and (b) devising the control logic to alternate between the two impedances.

To generate ∞ and 0 impedances, we can use a transistor to regulate the resistance between the gate and source ports of the transistor (denoted as R_{DS}), making it to be either ∞ or 0. This can be achieved by controlling the voltage between the gate and source ports (denoted as V_{GS}), where $V_{GS} < V_{th}$ for a duration of T_{ON} and $V_{GS} > V_{th}$ for a duration of T_{OFF} . V_{th} is the threshold voltage. Given that $T_s = T_{ON} + T_{OFF}$ is the sampling period T_s , we can determine T_{ON} and T_{OFF} for any desired S_{11} as follows:

$$\begin{cases} T_{ON} &= \frac{1}{2}(1 + S_{11})T_s, \\ T_{OFF} &= \frac{1}{2}(1 - S_{11})T_s. \end{cases} \quad (7)$$

Once the source port is connected to the ground, the first phase is complete.

In the second phase, we need to perform a high-frequency control of V_{GS} , by flipping the switch logic between $S_{11} = -1$ ($Z_L = 0$) and $S_{11} = 1$ ($Z_L = \infty$). This frequency must be several times higher than frequency that the LoRa receiver samples S_{11} , so that a high granularity and nearly continuous S_{11} can be produced. For instance, when a LoRa receiver samples the wireless signal at a frequency of f_s , the switching rate f_{sw} must satisfy $f_{sw} = N_{os} \cdot f_s$, if we need to generate N_{os} distinct quantization stages. Here, N_{os} denotes the oversampling factor. As shown in Fig. 4, by controlling the percentage of the ON or OFF state in the N_{os} oversampling slots, we can generate N_{os} distinct quantization stages, thereby producing N_{os} different values of S_{11} . This process can be considered as a passive digital-to-analog conversion, with a resolution of $\log_2(N_{os})$ bits.

C. IQ Mod. for Complex S_{11} Synthesis

So far we have demonstrated how to generate an arbitrary real-valued S_{11} using PWM. The next step is to generate arbitrary complex S_{11} values. This is essentially a problem of in-phase and quadrature (IQ) modulation in backscatter, which has been well addressed using Wilkinson power divider/combiner. In this section, we first briefly visit this

method, and then we present how to leverage it to synthesize an arbitrary complex-valued S_{11} .

Fig. 5 illustrates a schematic of the Wilkinson power divider/combiner, a three-port RF component whose operation consists of two stages:

In the first stage, port A receives the incident signal and equally divides it into two paths (port B and port C). Assuming the incident signal is $S_{in}(t) = A\cos(2\pi ft)$, the output signal at port B and port C is $y_{in,B}(t) = y_{in,C}(t) = 0.707 \cdot A\cos(2\pi ft)$. These two signals propagate along two separate paths. On one of the paths (e.g., the upper path in Fig. 5), the signal undergoes a $\pi/4$ phase shift before reaching the transistor (or RF switch). The transistor then reflects the signal with a reflection coefficient $\Gamma_B(t)$. The backscattered signal can be expressed as $0.707 \cdot \Gamma_B(t) \cdot A\cos(2\pi ft + \pi/4)$. The signal on the other path is directly reflected by the transistor impedance. The backscattered signal can be expressed as $0.707 \cdot \Gamma_C(t) \cdot A\cos(2\pi ft)$.

In the second stage, the transistors' reflected signals propagate to ports B and C of the Wilkinson power divider/combiner, respectively. Signal that arrives at Port B undergoes another $\pi/4$ phase shift, which can be expressed as: $y_{out,B}(t) = 0.707 \cdot \Gamma_B(t) \cdot A\cos(2\pi ft + \pi/2)$. While, signal that directly arrives at port C can be expressed as $y_{out,C}(t) = 0.707 \cdot \Gamma_C(t) \cdot A\cos(2\pi ft)$. Clearly, $y_{out,B}(t)$ experiences an additional $\pi/2$ delay with respect to $y_{out,C}(t)$. Consequently, these two signals form the in-phase and quadrature component of backscatter signals.

Then the Wilkinson power divider/combiner combines these two signals with a 3dB insertion loss: $S_{out}(t) = 0.707 \cdot (y_{out,B}(t) + y_{out,C}(t))$. Then, by denoting $\Gamma_I(t) = \Gamma_B(t)$ and $\Gamma_Q(t) = \Gamma_C(t)$, the output signal can be expressed as:

$$\begin{aligned} S_{out}(t) &= -0.5A\Gamma_I(t)\sin(2\pi ft) + 0.5A\Gamma_Q(t)\cos(2\pi ft) \\ &= \text{Re}\left\{\frac{A}{2}e^{-\pi/2}(\Gamma_I(t)\cos(2\pi ft) - \Gamma_Q(t)\sin(2\pi ft))\right\}, \end{aligned}$$

where Γ_I and Γ_Q are the reflection coefficients of in-phase path and quadrature paths, respectively. This form aligns with the standard IQ modulation form: $y(t) = I(t)\cos(2\pi ft) - Q(t)\sin(2\pi ft)$.

After the above two stages, the S_{11} value of the port A can be written as: $S_{11} = 0.5 \cdot (\Gamma_I(t) + j \cdot \Gamma_Q(t))$. Now, to perform IQ modulation, we adjust $\Gamma_I(t)$ and $\Gamma_Q(t)$ respectively, by controlling the corresponding transistors with PWM modulation signals. Then the final $S_{11} = x + j \cdot y$ can be an arbitrary value within the rectangular region: $\{(x, y) \mid |x| < 0.5, |y| < 0.5\}$ of the complex plane.

D. Validation with VNA

In order to validate the proposed design, we implement a prototype of the designed IQ modulator circuit and connect it to a vector network analyzer (VNA), which is a dedicated microwave measurement instrument used for assessing S-parameters. To synthesize arbitrary S_{11} values, we use FPGA to generate PWM signals for the I and Q paths, respectively, as shown in Fig. 6. The real value measured in a single RF switch is presented in Fig. 6(a), while Fig. 6(b) shows complex-valued S_{11} generated with 3-bit quantization. As can be seen, the complex-valued S_{11} form an ellipse rather than a circle,

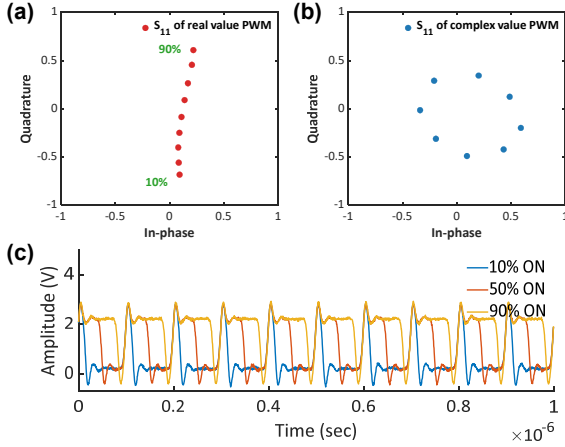


Fig. 6: (a) Real-valued S_{11} synthesis result using single RF switch. (b) Complex-valued S_{11} synthesis result using IQ modulator and PWM. (c) Switching waveform with different ON-OFF ratio.

which is caused by the slightly different insertion loss between I path and Q path, and the duty cycle rate is marked near their corresponding S_{11} points in the complex plane. The sampling rate of the oversampling modulator is 1 MHz. As can be seen, the oversampling modulation, together with the IQ modulator, can achieve a modulation granularity comparable with a complicated impedance loads array.

IV. CORE DESIGNS OF OSTAG

Although the aforementioned analysis suggests the feasibility of replacing impedance loads array using oversampling, there remain two challenges that must be addressed in order to ensure successful implementation.

A. Practical Design Challenges

C1: How to reduce the approximation error in PWM modulation. The oversampling PWM technique employed in synthesizing LoRa signals is inherently imperfect. A key aspect overlooked in previous analysis is the presence of an independent local sampling clock in the LoRa receiver, which differs from the tag's clock. This discrepancy can lead to a sampling time offset (STO), causing misalignment between the integral interval of the receiver filter and the PWM modulation period. This effect causes the approximation error in practical deployment. Furthermore, this error transforms into another form of noise, referred to as quantization error/noise in the subsequent text, which deteriorates the LoRa backscatter signal-to-noise ratio (SNR).

C2: How to address the clock imperfection of low-power oscillators. The core principle of our oversampling method involves trading clock speed for quantization resolution. It can be deduced that a 4-bit resolution necessitates a quantization clock rate that is sixteen times greater than the bandwidth (BW). Assuming a $BW = 500$ kHz, the resulting oversampling clock rate is 8 MHz, which significantly exceeds the clock rate of existing LoRa backscatter systems (3 MHz). Attaining such a high-frequency clock while maintaining power consumption below the mW level typically demands

the utilization of ring oscillators [40]–[43]. Nonetheless, ring oscillators exhibit challenges such as imprecise frequency stability across temperature variations and edge-to-edge period fluctuations (jitter). These limitations adversely affect the chirp SNR, consequently reducing the likelihood of correct LoRa reception.

B. S1: Delta-sigma Modulation for Chirps

We introduce a novel oversampling technique termed delta-sigma ($\Delta - \Sigma$) modulation, as a replacement for the conventional naive PWM oversampling. This approach uniformly disperses the ON pulse across the temporal domain utilizing a digital feedback framework. Consequently, the oversampling process dynamically selects ON or OFF state outputs, facilitating accurate reception of the LoRa signal from the tag without necessitating sampling clock alignment.

1) *Analyzing Quantization Error:* To analyze quantization error, we first write the tag signal $S_{11}(t)$ as follows:

$$S_{11}(t) = c(t) + q(t), \quad (8)$$

where $c(t)$ is the ideal LoRa chirp signal, and $q(t)$ is quantization error due to our oversampling method. After performing Fourier transform on both sides of Equ. 8, it becomes:

$$S_{11}(f) = C(f) + Q(f), \quad (9)$$

where $S_{11}(f)$ is the Fourier transform of the tag reflection signal $S_{11}(t)$, $C(f)$ is the Fourier transform of the ideal chirp $c(t)$, and $Q(f)$ is the Fourier transform of the quantization error $q(t)$. This equation indicates that the spectrum of the tag reflection signal is the superposition of the clean chirp spectrum, $C(f)$, and the quantization noise spectrum, $Q(f)$. Here, $Q(f)$, the Fourier transform of the quantization noise, can be expressed as:

$$\begin{aligned} Q(f) &= \int_{-\infty}^{+\infty} q(t) \cdot e^{-j2\pi ft} \cdot dt \\ &= \sum_{i=0}^{N_{sym}-1} \int_{iT}^{(i+1)T} q_i(t - iT) \cdot e^{-j2\pi ft} \cdot dt, \end{aligned} \quad (10)$$

where i is the index of the i^{th} LoRa symbol, N_{sym} is the number of the total symbols, T is the duration of a chirp signal: $T = 2^{SF}/BW$, and $q_i(t)$ is the quantization error in the i^{th} symbol. Now we can simulate the quantization error with the following steps. **Step1:** use the Monte-Carlo method to sample random LoRa chirps (symbols). **Step2:** obtain their corresponding $q_i(t)$ according to the approximation method (e.g. PWM). **Step3:** estimate the distribution of $q_i(t)$ based on Equ. 10 and thus obtain $\hat{Q}(f)$.

2) *Quantization Noise Spectrum of PWM:* Now we present the quantization noise spectrum $Q(f)$ using the method mentioned above and reveal why PWM method exhibits a unsatisfying noise spectrum. The $|C(f)|$ and $|Q(f)|$ are obtained under $SF = 10$, $BW = 500$ kHz, and oversampling factor 16 in Fig. 7. The PWM oversampling algorithm approximates the ideal LoRa spectrum $|C(f)|$, while the most of quantization error falls within the LoRa bandwidth, which is harmful for the passband LoRa demodulation. This effect can be further illustrated using spectrogram as shown in Fig. 8. Compared to the ideal chirp and 3-bit chirp using discrete loads, the

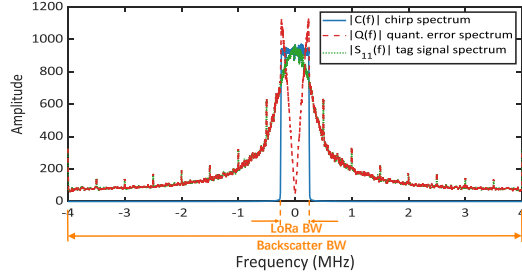


Fig. 7: Spectrum of ideal LoRa chirps, PWM quantization noise, and tag signal.

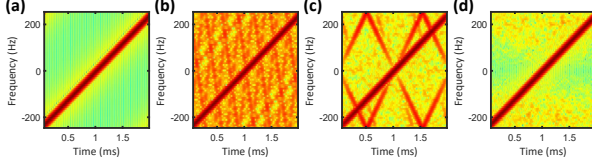


Fig. 8: Spectrogram of (a) ideal LoRa chirp, (b) 3-bit discrete loads, (c) naive $8\times$ oversampling using PWM, (d) $8\times$ oversampling using proposed $\Delta - \Sigma$.

PWM noise exhibits high power spectral density in an unwanted harmonic pattern. This harmonic pattern causes strong interference to a LoRa receiver. In summary, both the noise spectrum and spectrogram exposes the weakness of the naive PWM oversampling method.

The solution to this issue starts with the power characteristic of the oversampling process. We note that the total power of quantization error and perfect chirp $|S_{11}(f)|$ is power-constrained in frequency domain:

$$\int |S_{11}(f)|^2 \cdot df = C, \quad (11)$$

where C is a constant that satisfies $C = \alpha^2$ and α is the total insertion loss in the backscatter tag RF frontend; and for ideal RF frontend with IQ modulator, $C = 0.5$. Considering Equ. 9, if we squeeze the quantization error to the spectrum outside of the LoRa BW as shown in Fig. 7, the in-band LoRa backscatter signal would have higher perceived SNR. Higher SNR can make the backscatter signal more close to the ideal LoRa chirps and thus can lead to better performance.

3) *Noise Shaping using Delta-sigma Modulation*: Now our object is to minimize quantization noise in the LoRa bandwidth. This is realized by employing a technique known as noise shaping, which is commonly utilized in the design of high-performance analog-to-digital converter (ADC) circuits and digital signal processing. The fundamental principle of this technique is to mitigate the perceived noise in a signal by relocating it to the frequency range outside the region of interest using delta-sigma modulation technique [4], [5].

The structure of our chirp-modulator version is illustrated in Fig. 9(a). Initially, the input analog signal v_{in} undergoes comparison with the ultimate output result, followed by an integration process denoted as $A(z)$. Subsequently, the signal is converted to binary v_{out} through quantization. This quantization stage introduces the quantization error v_Q . With

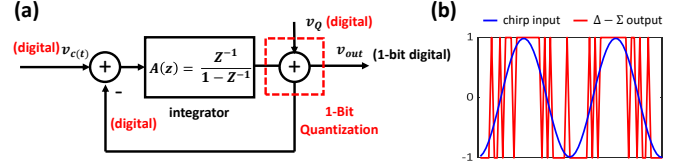


Fig. 9: (a) Design of the all-digital $\Delta - \Sigma$ modulator. (b) Chirp input and binary modulator output.

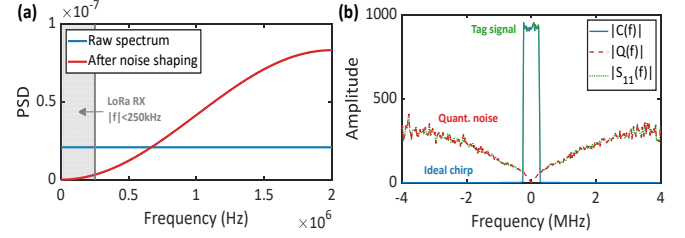


Fig. 10: (a) The noise spectrum before and after noise shaping. (b) The spectrum of ideal LoRa signal, quantization noise of $\Delta - \Sigma$ modulator, and tag signal after noise shaping.

integration of feedback, a significant portion of the power associated with v_Q is effectively shifted beyond the chirp bandwidth (BW), rendering it imperceptible to the receiver.

The time-domain proof is shown below:

$$v_{out}(nT_s) = v_{in}((n-1)T_s) + v_Q(nT_s) - v_Q((n-1)T_s). \quad (12)$$

After Z transform on both sides, we can obtain

$$v_{out}(z) = z^{-1}v_{in}(z) + (1 - z^{-1})v_Q(z). \quad (13)$$

It means that the final spectrum of the v_{out} contains two parts: the first part is from v_{in} and the other is from quantization noise v_Q . Here $(1 - z^{-1})v_Q(z)$ is the spectrum after noise shaping effect of the delta-sigma modulator. Next, we will illustrate the quantization noise spectrum and noise shaping result.

We first present the spectrum of quantization noise v_Q . The quantization error can be seen as a random variable Q that falls in the range of $[-1, 1]$ after one-bit quantization of S_{11} . Following Bennit's criteria, the random variable Q follows the probability density function $\rho(Q) = 1$, for $-1 \leq Q \leq 1$. Thus, we can obtain the total power of quantization noise P_Q :

$$P_Q = \int_{-1}^1 \rho \cdot Q^2 \cdot dQ = \frac{1}{3}. \quad (14)$$

Since $P(Q) = 2 \int_0^{f_s/2} V_Q(f)^2 \cdot df$ in frequency domain, the spectrum of quantization noise $V_Q(f)$ is derived:

$$V_Q(f) = \frac{1}{\sqrt{3f_s}}, \quad 0 \leq f \leq f_s/2. \quad (15)$$

Second, we show the spectrum after noise shaping. From Equ. 13, we replace z by $e^{j\omega}$, where ω is the normalized frequency $\omega = 2\pi \cdot f/f_s$, then we can obtain the new noise spectrum $V'_Q(f)$: $V'_Q(f) = (1 - e^{-j2\pi f/f_s}) \cdot \frac{1}{\sqrt{3f_s}}$.

Now we can obtain the single-sided power spectral density (PSD) of the noise spectrum before and after noise shaping based on $V_Q(f)$ and $V'_Q(f)$:

$$\begin{cases} PSD_Q(f) &= |V_Q(f)|^2 = \frac{1}{3f_s} \\ PSD'_Q(f) &= |V'_Q(f)|^2 = 4\sin^2(\pi \frac{f}{f_s}) \cdot \frac{1}{3f_s}. \end{cases} \quad (16)$$

We illustrate the PSD result in Fig. 10(a). In the figure, we suppose $BW = 500$ kHz and the oversampling factor is 8, i.e. $f_s = 8 \times BW$. Then we can see the quantization noise spectrum changed from uniform distribution to the S-shaped distribution where low-frequency range has significantly smaller noise power. This comparison demonstrates the benefit of noise shaping of delta-sigma modulator: the quantization noise is compressed in LoRa BW and enhanced in the outside frequency range. After the bandpass filtering of LoRa receiver, a high-quality LoRa signal would be received. This is further verified in our spectrum simulation as shown in Fig. 10(b). In contrast to Fig. 7, the quantization noise spectrum $|Q(f)|$ is significantly reduced in the LoRa receiving frequency range $|f| < 250$ kHz. The spectrogram improvement is illustrated in Fig. 8(d).

4) *SNR Increment with Oversampling Factor*: Now we investigate how oversampling factor can affect the SNR in the above delta-sigma modulation process. The backscatter modulation SNR can borrow the conclusion from the analog-to-digital conversion theory [4]:

$$SNR = 6.02M + 1.76 + 30\log_{10}N_{os}, \quad (17)$$

where N_{os} is the oversampling factor and M is the physical quantization resolution in bits (M bits means 2^M reflection coefficient levels). This equation shows that both the physical quantization levels and the oversampling factor can contribute to the improvement of backscatter SNR: (a) Doubling N_{os} can increase 9 dB SNR; (b) Doubling quantization levels (adding M by 1) can increase 6 dB SNR.

Thus, we can obtain the equivalent incremental quantization bits ΔM caused by the oversampling factor N_{os} :

$$\Delta M = \frac{30\log_{10}N_{os}}{6.02} = 1.5\log_2N_{os}. \quad (18)$$

Different from the previous expectation in Sec. III, every $2\times$ oversampling increases 1.5 extra quantization bits.

Remarks: Based on the analysis above, we can find that the clock frequency required to generated 3-bit oversampling requires the oversampling factor to be $N_{os} = 16$. When $BW = 500$ kHz, the final f_s should be 8 MHz. Typical circuits to realize μW power MHz-level oscillator is ring oscillator. The well-known drawbacks of this oscillator is the inaccuracy in center frequency and the instability in clock period. How to address this issue is the next challenge.

C. S2: Hybrid Clock Baseband Framework

In response to the challenges posed by low-power oscillators, we introduce a hybrid clock circuit framework. In essence, our approach involves utilizing a high-frequency, free-running ring oscillator exclusively for driving the oversampling logic to create a virtual impedance. Concurrently, the tasks of frequency shift and CSS modulation are executed by a precise, low-frequency crystal-based oscillator. Consequently, any frequency deviations within the ring oscillator primarily manifest as variations in quantization resolution.

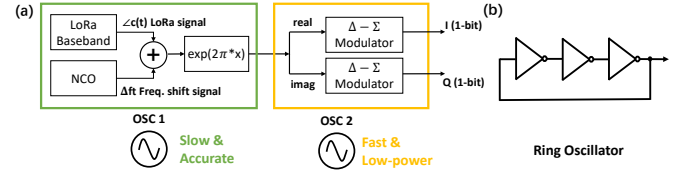


Fig. 11: (a) OsTAG baseband framework and its clock domains. (b) Ring oscillator design for the fast clock domain.

1) *Issue of Single-clock Framework*: Frequency shift-based backscatter tags typically utilize only one clock source from the analog circuits and divide the frequency into two parts. The first part is the baseband clock that controls the symbol synthesis and determines which symbol to send. The second clock generates the shift frequency. However, this framework cannot be directly applied to the oversampling tag, as the high-frequency oversampling clock results in a high frequency offset. This frequency offset propagates to the baseband clock after frequency divisions, potentially causing CFO and SFO.

2) *Design of Clock Domains*: To address this issue, OsTAG employs a hybrid-clock framework for replacement. Two analog circuits generate two distinct frequencies, resulting in the creation of two independent clock domains. The first clock domain is driven by the slow clock, a crystal-based circuit providing an accurate frequency. This low-frequency, stable, and precise oscillator generates valid LoRa baseband signals and shift frequencies, minimizing CFO and SFO. The second clock domain is driven by the fast clock, utilizing an inaccurate and unstable low-power ring oscillator to drive the delta-sigma oversampling modulator alone.

The circuit structure and its clock domains are illustrated in Fig. 11. The first clock domain enables digital domain precise generation of $y(t) = c(t) \cdot e^{j2\pi\Delta f t}$, where Δf is the shift frequency for avoiding self-interference with the single-tone transmitter. The second clock domain contains the oversampling modulator. In the OsTAG design, $\Delta f = 2$ MHz, and the first clock domain adopts a clock frequency of 2 MHz which drives frequency shift logic and the baseband after frequency division. The fast clock employs 16, 32, and 64 MHz frequencies to attain different oversampling factors over baseband bandwidth (BW).

3) *Impacts of Clock Imperfections*: In this section, we analyze the impacts of clock imperfections on our design. We assume that the first clock domain is accurate, and thus the $y(t)$ signal, which becomes the v_{in} signal in Fig. 9, is ideally generated. We then examine how the frequency offset affects the oversampling modulation.

First, we investigate the frequency offset of the average oversampling frequency. From Equ. 16, we observe that the offset of f_s alters the PSD of the quantization noise, subsequently changing the total perceived quantization noise within the LoRa bandwidth:

$$\begin{aligned} P_q &= \int_0^{BW/2} 4\sin^2\left(\pi\frac{f}{f_s}\right) \cdot \frac{1}{3f_s} \cdot df \\ &= \frac{BW}{3f_s} - \frac{1}{3\pi} \cdot \sin\left(\frac{\pi BW}{f_s}\right), \end{aligned} \quad (19)$$

where P_q is the power of quantization noise in the LoRa band, and BW is the bandwidth. Since BW is fixed in a packet, it

becomes evident that when $\Delta f_s/f_s$ is small, P_q decreases as f_s increases because $\frac{dP_q}{df_s} < 0$. Therefore, if the oscillator generates an oversampling clock with a center frequency larger than the expected f_{s0} , it benefits LoRa backscatter. Conversely, if the clock's center frequency is smaller than the expected f_{s0} , it reduces SNR.

Second, we analyze the impacts of clock jitter on f_s . It is important to note that clock jitter essentially represents the variation in clock periods. This variation causes the sampling points to deviate from the ideal time. Considering the $y(t)$ written as a function of instantaneous frequency $f(t)$: $y(t) = \cos(2\pi f(t)t)$, the slew rate of $y(t)$ at any moment can be obtained:

$$\frac{dy(t)}{dt} = -2\pi f(t) \cdot \sin(2\pi f(t)). \quad (20)$$

The jitter can now be related to the quantization error. Let the jitter be ΔT_s , and obtain:

$$\left| \frac{\Delta y(t)}{\Delta T_s} \right| \approx \left| \frac{dy(t)}{dt} \right| = |2\pi f(t) \cdot \sin(2\pi f(t))| \leq 2\pi |f(t)| \quad (21)$$

The maximum $|f(t)|$ is $f_s/2$. To avoid degrading the quantization performance, a safe method ensures $\Delta y(t) < \frac{1}{2^N}$. This implies that ΔT_s should satisfy $\Delta T_s \leq \frac{1}{\pi 2^N} \cdot \frac{1}{f_s}$. Conversely, the maximum quantization bits loss can be estimated using the equation: $\Delta N \approx N - \log_2(\pi \frac{\Delta T_s}{T_s})$. The OsTAG aims for an oversampling factor of $32 \sim 128$, so the jitter only requires 0.5%-1.0%, which is easily achievable in oscillator designs.

Remarks: We can find that both center frequency offset Δf_s and clock jitter ΔT_s only impact quantization noise and do not cause CFO or SFO.

4) *Choice of oversampling factor:* Now we consider how to choose the appropriate oversampling factor in OsTAG deployment, which is essentially a trade-off between performance and power. From Equation 17, we can find that as long as we can increase the oversampling factor by enhancing the frequency of the fast clock circuit for delta-sigma modulation, the backscatter SNR can always increase, thereby achieving better communication performance. On the other hand, the dynamic power will correspondingly increase in the circuit linear to operating frequency when the oversampling factor increases. In Figure 12, we illustrate the relation between the oversampling factor, quantization SNR, and the power consumption, with the power consumption details would be given in Section VI-F.

Basically, the deployment of OsTAG can choose the maximum oversampling factor if the power consumption is within the power budget, which can be further obtained either from an external RF energy harvesting circuit or the capacity and lifespan of a battery. The oversampling clock, which is the all-digital ring oscillator as shown in Fig. 11, can be programmed to different frequencies by adjusting the in-loop delay.

V. IMPLEMENTATION

A. Hardware Prototype

We implement the OsTAG prototype to verify and evaluate our design, realizing the whole circuit on a PCB. The all-digital baseband is written in Verilog and deployed on a low-power FPGA GW1N-4B from GOWIN. The adoption

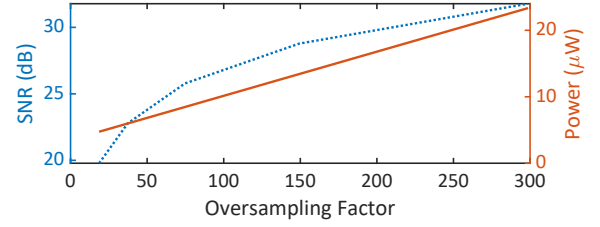


Fig. 12: The relation between the oversampling rate the backscatter signal SNR and power consumption.

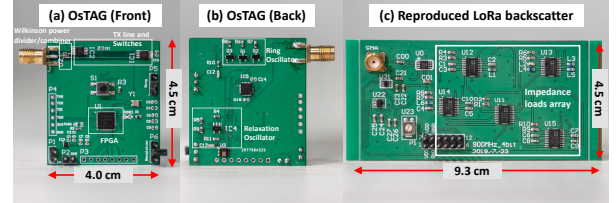


Fig. 13: (a) OsTAG hardware prototype front, (b) back, (c) the reproduced existing LoRa backscatter (with inductors).

of FPGA realizes real-time baseband signal streaming to the I/O. It consumes around 3.3 mW, which can be reduced to μ W after IC design as to be introduced in Section V-C. The analog frontend consists of two RF switches and transmission lines that create a $\pi/4$ phase difference. Other analog circuits include the LDO for voltage regulation and the ring oscillator. The two oscillators can be selected to provide the oversampling clock frequency. The RF switches are ADG902, and the transmission lines are realized by controlling the track width W under the constraint of PCB board thickness H . The ring oscillator inverters are realized with N-channel MOSFET BSS138P, and the delay between inverters is controlled by the drain resistance. Notably, we write the entire LoRa modem in Verilog in our digital baseband, basing it on and verifying it with the open-access MATLAB code LoRaPHY [37] for valid LoRa modulation. For comparison, we also reproduce the circuit in the existing LoRa backscatter, which is made of two-stage cascading ADG904 4:1 multiplexers.

The OsTAG prototype is depicted in Fig. 13(a)(b), and the reproduced LoRa backscatter appears in Fig. 13(c). It is evident that the removal of the impedance loads array significantly reduces the size of the OsTAG PCB circuit compared to the existing LoRa backscatter circuit. Comparing only the RF frontend, the OsTAG design occupies 1.75 mm^2 , while the impedance loads array takes up 19.25 mm^2 of the circuit area. This results in a 90.9% reduction in area. The circuit area advantage of oversampling when implemented in an IC is reported in Section VI-F.

B. Transmitters and Receivers

Similar to the existing LoRa backscatter, the transmitter is a single-tone transmitter. Various methods can be employed to implement such a transmitter, including using an RF frequency synthesizer cascaded to a power amplifier (PA), utilizing a plugged-in analog signal generator, or simply adopting software-defined radio. For indoor experiments, a Keysight N5171B with 30 dBm transmitter power is used, while a

USRP B210 from Ettus Research is employed for outdoor experiments due to its portability. Powered by USB, it has a maximum transmission power of approximately 19dBm. The GNURADIO software with *USRP sink* block is used to access it. As for the receiver, the commodity device SX1276DVK1KAS [30] serves as the development kit for the SX1276 LoRa chip from Semtech.

C. IC Design

The baseband of OsTAG is synthesized and laid out using the industry-standard IC design tool, Synopsys. Employing the SMIC 40 nm 1.1 V ULP (Ultra-Low Power) process, the IC area and power consumption of the OsTAG design can be estimated. All analog IC circuits, including the RF switch, LDO, temperature-compensated crystal oscillator (TCXO), and ring oscillators, are designed using the industry-standard software, Cadence. The power consumption and area occupation are presented in Sec. VI-F.

VI. EVALUATIONS

A. LoRa Packet Spectrum

We first examine the final LoRa spectrum of the OsTAG of our oversampling modulator using delta-sigma modulation. The known advantage of existing 3-bit LoRa backscatter lies in its harmonic and image cancellation capabilities. To compare, three types of tags are set in the same test environment, and their spectrum performance is measured. (a) Tag1: The reproduced LoRa backscatter using only 1 bit, switching between two impedance loads. These loads correspond to the S_{11} values e^{j0} and $e^{j\pi}$. (b) Tag2: The reproduced LoRa backscatter using 3 bits, switching among 8 impedance loads. (c) Tag3: The OsTAG performs delta-sigma oversampling to generate a LoRa packet. The excitation source is an analog signal generator that sends at 20dBm; and the spectrum analyzer (Keysight N9322C) measures their spectrum one meter away. The tags are placed in the midpoint respectively. All the tags are configured to send $SF = 10$ and $BW = 500\text{kHz}$ LoRa packet.

The spectrum results are shown in Fig. 14. It is observed that the 1-bit LoRa backscatter signal is double-sideband and contains 3rd and 5th harmonics, which can challenge LoRa transmission in other channels. In contrast, the OsTAG, with 1-bit impedance in both I and Q paths, successfully cancels all these images and harmonics. The 3rd harmonic is compressed by 21dB, the 5th harmonic by 13dB, and the image by 37dB. The 5th harmonic itself is too small to accurately evaluate the cancellation performance, so the actual performance of 5th harmonic cancellation should be well above 13dB. These results demonstrate that the OsTAG achieves nearly the same performance as the existing LoRa backscatter [31], which reports 22dB compression for the 3rd harmonic and 14dB compression for the 5th harmonic.

B. Indoor Experiments

Indoor measurements are conducted in one layer of a school building. The excitation signal source is placed at the

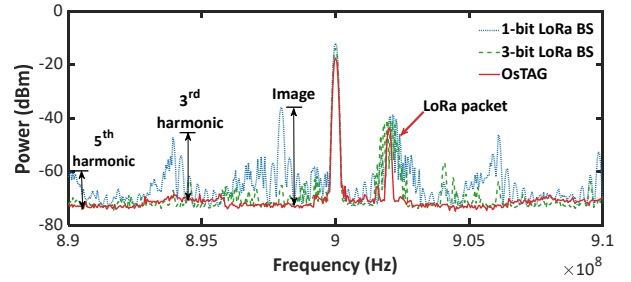


Fig. 14: The LoRa signal spectrum comparison among 1-bit, 3-bit LoRa backscatter, and OsTAG.



Fig. 15: Schematic representation of the indoor experimental environment.

corner of an office room, transmitting a 30dBm single-tone signal at 900 MHz. Tags are placed in different office rooms with a SX1276 receiver nearby (1 meter away) in the same room/corridor. All tags transmit LoRa packets with $SF = 10$ and $BW = 500\text{kHz}$. In total, 6 different office rooms are explored, and successful decoding of the LoRa packet occurs in all these rooms. The RSSI values at each position are recorded, with the results and locations of the single-tone transmitter, receiver, and tags shown in Fig. 15. It is important to note that all the walls in the figure are made of concrete.

C. Outdoor Experiments

The OsTAG system is also evaluated in outdoor environments. Figure 16 illustrates two deployments of the OsTAG system. Deployment 1 is situated near the school building, while Deployment 2 is close to a large lawn on the campus. The range between the transmitter (USRP B210 at 19dBm) and the commodity LoRa receiver is fixed to 100 m. The Tx-tag distance d_1 is swept while the tag-Rx distance d_2 changes accordingly by following $d_2 = 100\text{m} - d_1$. The RSSI value of the received LoRa backscatter packets is measured and logged using SX1296KA software. Figure 17 demonstrates the relationship between LoRa packet RSSI values and d_1 . Points with -130dBm RSSI indicate that packet loss exceeds 90% at that location.

D. BER and Throughput Comparison with Existing Work

Now we compare the BER and throughput performance of our OsTAG to that of the LoRa backscatter. The LoRa

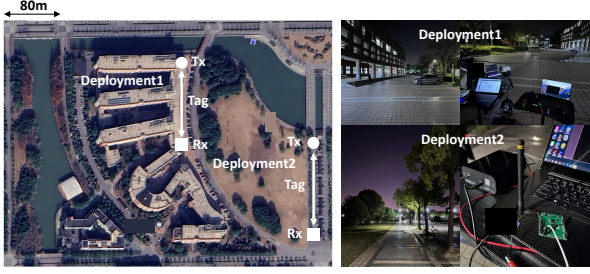
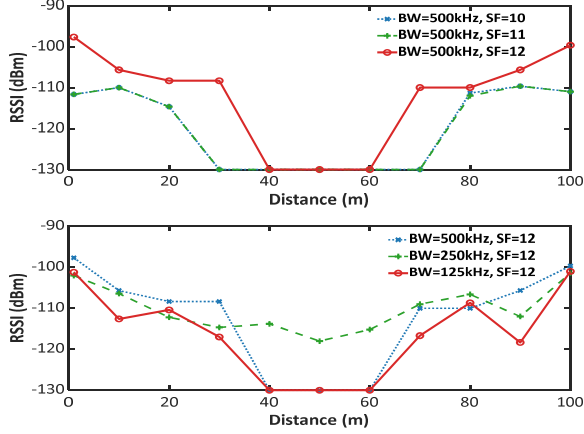


Fig. 16: The outdoor deployment of OsTAG in the campus.

Fig. 17: The RSSI values at different Tx-tag distances (d_1).

backscatter tag is reproduced according to the literature, as shown in Fig.13. We conduct two experiments: (a) The first experiment is conducted indoor, with Tx-tag distance fixed to $d_1 = 1$ m and sweeping the tag-Rx distance from 5 m to 100 m; (b) The other experiment is conducted outdoor with Tx-tag distance $d_1 = 1$ m and tag-Rx distance sweeping from 100 m to 1.2 km, with SF fixed to 12. In both experiments, the BW is set to 125 kHz.

The indoor experiment results are illustrated in Figure 19 and Figure 20. Note that zero BERs are changed into 1×10^{-3} to fit the log-scale axis. It can be seen that the OsTAG constantly realizes zero BER and high throughput in all locations while the existing LoRa backscatter suffers from high BER and throughput degradation when the tag-Rx distance exceeds 65 m. The outdoor experiment results are shown in Figure 21. It can be seen that OsTAG can still communicate with distance up to 1.2 km while existing work can only operate within 100 m. This is because OsTAG can achieve 7-bit quantization in backscatter modulation with the oversampling method while LoRa backscatter only achieves 3-bit quantization.

E. Concurrent Transmissions

LoRa supports FDMA-like concurrent transmission by utilizing different channels. In OsTAG, this can be achieved by performing various frequency shifts. The channel spacing is set to 2 MHz under $SF = 10$ and $BW = 500$ kHz. Two concurrent tags, tag A and tag B, are randomly placed in two indoor locations, 50 cm apart. The measurement results from these locations are displayed in Fig. 18. It demonstrates that OsTAG maintains similar performance when a concurrent tag

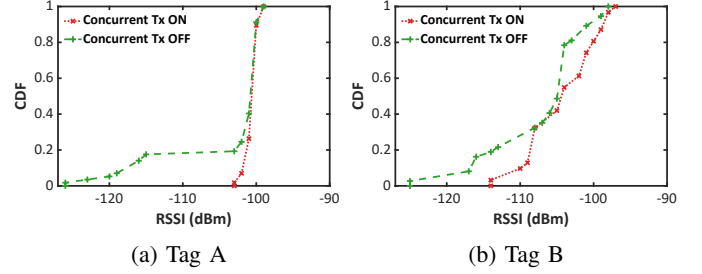


Fig. 18: Concurrent transmissions.

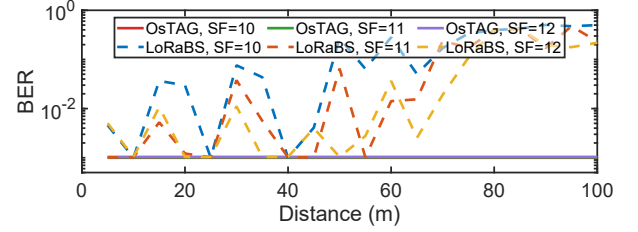


Fig. 19: Indoor BER comparison with existing work.

is in close proximity. We also evaluate the BER performance when the two tags are placed at distances ranging from 1 m to 5 m apart. As Figure 22 shows, the BERs are below 1×10^{-3} in measurements.

Concurrency scaling analysis: With the FDMA method, the concurrency of the system can continuously increase as long as the receiver RF front-end can support the total spectrum bandwidth of the tags:

$$N = \max \left\{ N \left| \sum_{i=1}^N BW_{\text{tag},i} \leq BW_{\text{Rx}} \right. \right\},$$

where $BW_{\text{tag},i}$ and BW_{Rx} are the LoRa BW of the i^{th} tag and the receiving bandwidth of Rx, respectively. We perform higher concurrency experiments to validate the performance scaling of OsTAG. We deploy 1-5 tags in the experiment and evaluate the FDMA concurrency performance under these conditions. The BER and throughput performance are illustrated in Figure 23, where zero BERs are converted to 10^{-4} . It further proves the scalability of concurrent transmissions. We note that parallel decoding of multiple LoRa packets in a single channel is also a research topic, which is mainly based on the Rx's non-standard signal demodulation and is beyond the scope of this paper.

F. IC Circuit Area and Power

IC circuit area. The circuit area of the IC is evaluated to highlight the core advantage of OsTAG. For the analog circuits, the LDO, TCXO, and RF switch occupy a total area of $1.3 \times 10^5 \mu\text{m}^2$. For the digital baseband, the LoRa baseband circuit occupies $2.3 \times 10^4 \mu\text{m}^2$, and the delta-sigma modulator circuit occupies $88 \mu\text{m}^2$. In total, OsTAG consumes $1.5 \times 10^5 \mu\text{m}^2$ of functional area. This size is comparable to the Alien Technology ALN-9640 passive RFID chip ($1.6 \times 10^5 \mu\text{m}^2$) [3].

A fair estimation of the existing LoRa backscatter circuit area is also provided. According to the literature, it requires the

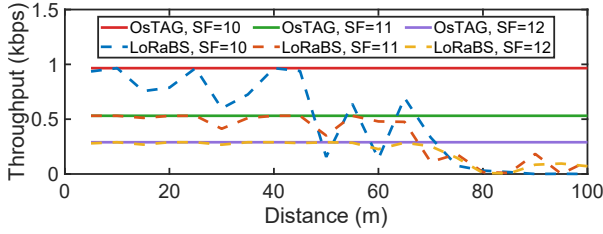


Fig. 20: Indoor throughput comparison with existing work.

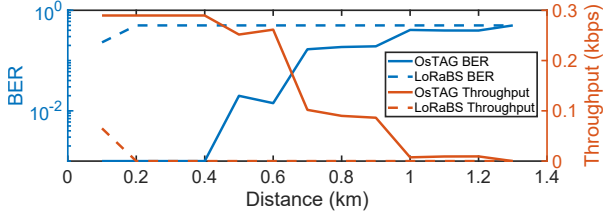


Fig. 21: Outdoor performance comparison with existing work.

LDO, TCXO, and RF switches similar to the OsTAG design. The main difference is the need for three large inductors as impedance loads in the circuit, estimated to consume $3.81 \times 10^5 \mu\text{m}^2$ as discussed at the end of Sec. II-B. Additionally, it needs a LoRa baseband to realize CSS modulation, which is similar to the OsTAG LoRa digital circuit ($2.3 \times 10^4 \mu\text{m}^2$). Finally, the hybrid analog-digital modulator circuit can be implemented with a streamlined numerically-controlled oscillator (NCO) and is negligible compared to the size of the LoRa baseband. In total, the existing LoRa backscatter is estimated to consume $5.3 \times 10^5 \mu\text{m}^2$. OsTAG achieves a 72% reduction in functional circuit area.

IC power consumption. The power consumption of OsTAG can be divided into several aspects. (a) For the analog circuit, the LDO takes $\sim 19 \text{ pW}$, the RF switch takes $0.4 \mu\text{W}$ to $1.3 \mu\text{W}$ under different oversampling rates, and the TCXO generating 2 MHz consumes $3.5 \mu\text{W}$. The ring oscillator is realized in a similar form with [6] and consumes $0.21 \mu\text{W MHz}^{-1}$; and it consumes $3.4 \mu\text{W}$ to $13.4 \mu\text{W}$ under 16 MHz to 64 MHz oversampling clock frequency. (b) For the digital baseband circuit, the LoRa modulation consumes $12 \mu\text{W}$ no matter it runs at 125 kHz or 500 kHz bandwidth because its leakage power is dominant. The oversampling modulator that generates arbitrary complex S_{11} values consumes $0.5 \mu\text{W}$ to $2.0 \mu\text{W}$ driven by the ring oscillator of 16 MHz to 64 MHz. To summarize, the OsTAG consumes $7.8 \mu\text{W}$ to $20.2 \mu\text{W}$ according to the oversampling clock.

This power consumption is comparable to the $9.25 \mu\text{W}$ reported in the literature [31]. When the oversampling clock is 16 MHz, the power consumption is even lower than the existing LoRa backscatter due to the elimination of DAC and fewer RF switches. When the oversampling rate is 64 MHz, the oversampling clock causes more power consumption yet brings higher quantization resolution.

G. Overall performance comparison

We compare the overall performance of the proposed oversampling work with a common LoRa backscatter tag from

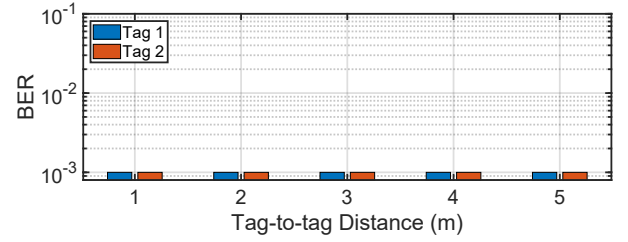


Fig. 22: BER of two concurrent tags with different tag-to-tag distance.

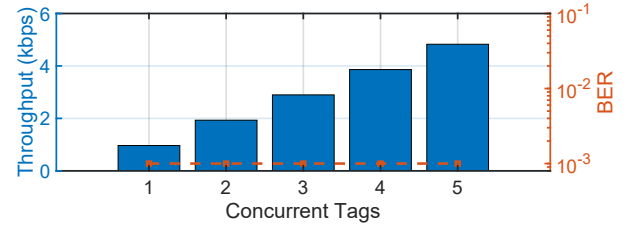


Fig. 23: BER and throughput of 1-5 concurrent tags.

the aspects of size, cost, energy consumption, throughput, and communication distance. The results are summarized in Table I. The comparison metrics include: communication distance, tag size, on-tag CSS, harmonic cancellation, single-tag throughput, and power consumption.

The metric ‘distance’ is presented as Tx-tag (d_1) and tag-Rx (d_2) distance pairs to provide better clarity. Tag size is categorized as either ‘large’ or ‘small,’ depending on whether it includes a complex inductor-based impedance network (infeasible/feasible for IC implementation). The metric ‘on-tag CSS’ indicates whether the tag performs CSS modulation, which simplifies the design of the transmitter by only requiring single-tone transmission. The metric ‘harmonic cancellation’ refers to the tag’s ability to suppress image signals, third-order harmonics, and fifth-order harmonics caused by quantization.

From the table, we observe that OsTAG is the only design capable of simultaneously achieving a small tag size, high-data-rate LoRa transmission (0.37 kbps $\sim$ 27.3 kbps, LoRa PHY compatible), $10 \mu\text{W}$ -level power consumption, and 1000 m-level long-range backscatter.

VII. DISCUSSIONS

■ **Why is placing impedance loads outside the chip not recommended?** There are three primary reasons. Firstly, doing so would necessitate the use of additional I/O pads, resulting in significant area consumption. Secondly, the bonding wire that connects the chip circuit and the loads introduces small yet non-negligible coupling capacitance and inductance. Thirdly, discrete components are not precise in their values. Due to these reasons, placing impedance loads inside the chip is a common choice, which brings the inductor challenge.

■ **Can oversampling be applied to other backscatter systems?** Yes, the $\Delta - \Sigma$ modulator design can be extended to other waveform synthesis methods, such as BLE GFSK or ZigBee signal generation. However, these protocols operate in the 2.4 GHz band, where the inductor size is much smaller

TABLE I: Performance comparison between OsTAG and other LoRa backscatter systems

Work	Distance (d_1, d_2)	Tag size	On-tag CSS	Harmonic cancellation	Single-tag throughput	Power consumption
LoRa Backscatter [31]	5 m, 2.8 km	Large	Y	Y	0.09 kbps	9.25 μ W
PLoRa [27]	1 m, 1 km	Small	N	N	0.1 kbps	220 μ W
MobiScatter [33]	~ 10 m, ~ 10 m	Large	Y	Y	1 kbps	62.6 μ W
P2LoRa [16]	1 m, 0.5 km	Small	N	N	0.11 kbps	320 μ W
OsTAG (this work)	1 m, 1.2 km	Small	Y	Y	0.37 kbps $\sim$ 27.3 kbps	7.8 μW to 20.2 μW

than in sub-GHz frequencies, resulting in reduced area benefits from oversampling. Our system can also be applied to concurrent backscatter systems, such as the Netscatter system [12]. Applying our method to this system can eliminate in-band quantization noise and thus reduce mutual interference among tags, helping further overcome the critical near-far problem. This will be our future work.

■ Why does OsTAG have a shorter communication range?

Compared to LoRa backscatter, OsTAG has a shorter communication range of 1.2 km between Tx and Rx, as opposed to the range reported in the literature [31]. This is due to two main factors. First, existing LoRa backscatter utilizes a non-standard, extremely narrow $BW = 31.25$ kHz that only exists in test mode, resulting in a few seconds' duration for a four-byte packet, while our design is based on standard LoRa setup with $BW = 125/250/500$ kHz. Second, the IQ modulator introduces 3dB inherent loss due to the Wilkinson divider/combiner, which can be translated to several hundred meters range reduction when SNR is low.

VIII. RELATED WORKS

LoRa backscatter. The LoRa backscatter system [31] enables backscatter communication over a hundred-meter range with a single-tone excitation source. In contrast, PLoRa [27] shifts the ambient LoRa signal in frequency for data piggy-backing but lacks support for commodity reception. Similarly, Aloha [11] also faces this drawback. RF transformer and SD-PHY backscatter [10], [44] provide general frameworks for achieving commodity LoRa reception. However, both LoRa backscatter and RF transformer are not IC-friendly. The former requires large-sized inductors, as our analysis in this paper demonstrates, while the latter employs GaAs pHEMT transistors that are incompatible with Si-based CMOS processes. SD-PHY uses vanilla ED to achieve synchronization with ambient LoRa transmitters over a limited range. In comparison, our design represents the first IC-efficient design (similar in size to an RFID) for commodity LoRa backscatter, achieving long-range communication.

Inductor-free designs. Inductor-free designs are crucial for reducing circuit size and fabrication complexity in areas like RF communication circuits and energy harvesting [7], [8], [21], [22], [32], [36], [45]. In contrast to OsTAG, which adopts an oversampling-based modulation to achieve inductor-free LoRa backscatter, these solutions typically rely on transistor-based active inductors to replace physical inductors or use capacitance-based designs, which are limited to energy harvesting applications. For μ W-level backscatter modulation, these solutions are not applicable.

Existing oversampling methods. In wireless systems such as RFID, ZigBee, and BLE, existing oversampling methods are

typically applied at the receiver for two main purposes: (a) To enable parallel decoding of concurrent packets, such as RFID parallel decoding [17], [18], [26] and ZigBee multi-packet demodulation [35]. In this application, oversampling provides multiple sampling points for a received symbol, enabling fine-grained time-domain information. (b) To achieve longer range. BLE receivers use oversampling and digital signal processing techniques to enhance receiver sensitivity, thus extending the typical 10 m range to 100 m or more [13]. In contrast, our oversampling method is applied to the low-power tag to enable inductor-free LoRa backscatter modulation.

IX. CONCLUSIONS

In this paper, we present OsTAG, a novel backscatter tag that utilizes delta-sigma modulation to achieve LoRa modulation without the need for a physical impedance loads array. By eliminating the need for inductors, the size of the LoRa backscatter IC circuit is significantly reduced. We have constructed a prototype of OsTAG to verify our design. Our results demonstrate that OsTAG can effectively cancel harmonics and achieve a communication range of several hundred meters, comparable to traditional methods.

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